Lesson 1: Solving Linear–Quadratic Systems

Finding where a line and a parabola meet, by graphing and by substitution or elimination, and predicting the number of solutions.

1. Linear–Quadratic Systems

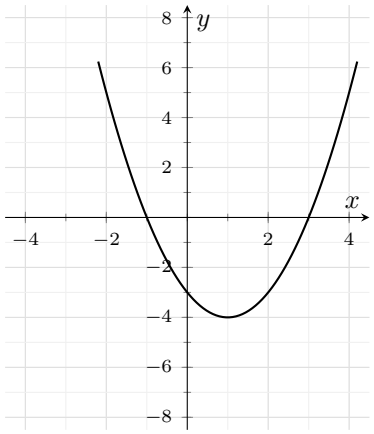
A **linear–quadratic system** is a pair of equations in x and y: one linear (its graph is a line, such as $y = mx + b$) and one quadratic (its graph is a parabola, such as $y = ax^2 + bx + c$). A **solution** of the system is an ordered pair (x, y) that makes **both** equations true, so the solutions are the **points of intersection** of the line and the parabola.

A line and a parabola can meet in **two points**, **one point** or **no points**, so the system has 2, 1 or 0 real solutions. A line that touches a parabola at exactly one point without crossing it is **tangent** to the parabola. (A vertical line $x = k$ also meets a parabola $y = ax^2 + bx + c$ exactly once, but it crosses it, so it is not a tangent.)

To **verify** a solution, substitute it into **each** original equation; both must be true.

Example 1

The parabola $y = x^2 - 2x - 3$ is graphed. For each line, graph the line on the grid, then state the number of solutions of the system and the solution(s).



(a) $y = -2x + 1$

(b) $y = -2x - 3$

(c) $y = -2x - 6$

(d) Verify the solution $(-2, 5)$ from part (a).

2. Solving by Graphing with a Calculator

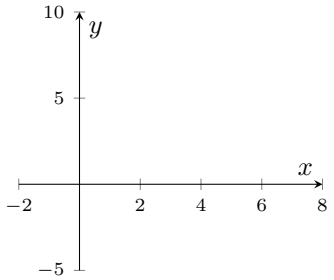
When the intersection points are not on grid lines, a graphing calculator finds them.

TI-84 Plus CE (the Casio has the same tool under G-Solv, ISCT):
1. Write each equation in the form $y = \ldots$ Enter the quadratic as Y_1 and the linear one as Y_2, then press GRAPH.
2. Adjust the WINDOW until every intersection point is on the screen.
3. Press 2nd [TRACE] (CALC), 5:intersect. Press ENTER for the first curve and for the second curve, move the cursor near one intersection point, and press ENTER for the guess.
4. Repeat step 3 near the other intersection point. Round only at the end, as the question asks.

A graph gives decimal approximations; an algebraic method (Sections 3 and 4) gives exact answers.

Example 2

Consider the system $y = x^2 - 4x + 1$ and $y = 2x - 3$.
(a) Use a graphing calculator to solve the system. Give each coordinate to the nearest tenth, and state a window that shows both points.



(b) Sketch what the calculator shows.

3. Solving by Substitution

Substitution: (1) write the linear equation as $y = \ldots$ (or $x = \ldots$); (2) substitute that expression into the quadratic equation; (3) write the resulting quadratic equation in standard form and solve it (factor, or use the quadratic formula); (4) substitute each x-value into the **linear** equation to find y; (5) verify in both equations and write each solution as an ordered pair.

Each x-value gives exactly one point. If the resulting equation has a double root, the line is tangent to the parabola.

Example 3

Solve each system algebraically.

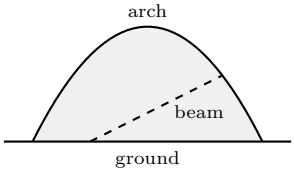
(a) $y = x^2 + 2x - 5$
$y = 3x + 1$

(b) $y = 2x^2 + x - 4$
$4x - y = 2$

(c) $y = x^2 - 4x + 1$
$y = 2x - 3$
(exact answers: Example 2)

Example 4

A tunnel entrance has the shape of the parabola $y = -0.25x^2 + 4$, where x and y are in metres and the ground is the x-axis. A straight support beam follows the line $y = 0.5x + 1$. Determine, to the nearest hundredth of a metre, the point where the beam meets the arch. Explain why one solution of the system is rejected.



4. Solving by Elimination

If both equations are written with y alone on one side (or with matching y-terms), subtracting one equation from the other **eliminates** y and leaves a quadratic equation in x. If the y-terms are opposites, add the equations instead. Then finish as in substitution: solve for x, find each y from the simpler equation, and verify.

Example 5

Solve each system by elimination.

(a) $y = -x^2 + 4x + 5$
$y = -2x + 10$

(b) $x^2 + y = 7$
$3x - y = -3$

5. The Number of Solutions and the Discriminant

Substitution turns the system into one quadratic equation $ax^2 + bx + c = 0$. Each real root gives one point of intersection, so the **discriminant** $b^2 - 4ac$ of that equation tells how many solutions the system has, without solving it.

Discriminant of the resulting equation	**Line and parabola**	**Solutions of the system**
$b^2 - 4ac > 0$	intersect at two points	two
$b^2 - 4ac = 0$	the line is tangent to the parabola	one
$b^2 - 4ac < 0$	do not intersect	none

Always write the resulting equation in standard form (one side 0) before reading a, b and c.

Example 6

Without solving, determine the number of solutions of each system.

(a) $y = 2x^2 - 3x + 1$
$y = x - 1$

(b) $y = x^2 - 4x + 7$
$y = 2x - 3$

(c) $y = -x^2 + 5$
$y = -3x + 2$

Example 7

(a) For which values of k does the line $y = 2x + k$ intersect the parabola $y = x^2 - 4x + 3$ at two points? at one point? at no points? For the one-point case, determine the point of tangency.

(b) The line $y = mx - 1$ is tangent to the parabola $y = x^2 + 3$. Determine the possible values of m.

6. A Line and a Circle

The same methods work for other curves. The equation $x^2 + y^2 = r^2$ describes a **circle** with centre $(0, 0)$ and radius r: by the Pythagorean theorem, every point (x, y) on it is r units from the origin. A line meets a circle at 2, 1 (a tangent) or 0 points. Substitute the line into the circle's equation, solve, then use the **line** to find y.

Example 8

(a) Determine the exact points of intersection of the circle $x^2 + y^2 = 20$ and the line $y = 2x$.

(b) Show that the line $x + 2y = 10$ is tangent to the circle $x^2 + y^2 = 20$, and determine the point of contact.

Lesson 2: Solving Quadratic–Quadratic Systems

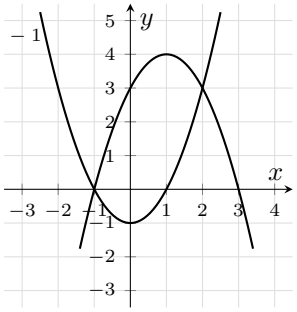
Finding where two parabolas meet, by graphing and by elimination or substitution, and recognizing systems with 0, 1, 2 or infinitely many solutions.

1. Quadratic–Quadratic Systems

A **quadratic–quadratic system** is a pair of quadratic equations, such as $y = a_1x^2 + b_1x + c_1$ and $y = a_2x^2 + b_2x + c_2$. Its solutions are the points of intersection of the two parabolas. Two different parabolas meet in **two points**, **one point** or **no points**. If the two equations describe the **same** parabola (they may look different, as $y = (x-1)^2$ and $y = x^2 - 2x + 1$ do), every point on it is a solution: the system has **infinitely many** solutions.

Example 1

The graphs of $y = x^2 - 1$ and $y = -x^2 + 2x + 3$ are shown.



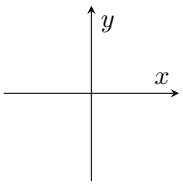
(a) State the solutions of the system.

(b) Verify each solution in both equations.

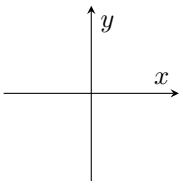
Example 2

On each set of axes, sketch two parabolas that form a system with the given number of solutions. Write a pair of equations for (d).

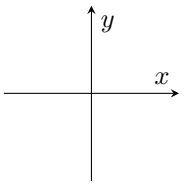
(a) no solution



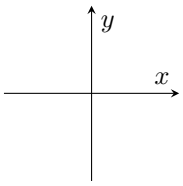
(b) one solution



(c) two solutions



(d) infinitely many



2. Solving Algebraically

Write both equations as $y = \ldots$ Setting the right sides equal (substitution) is the same as subtracting one equation from the other (elimination): y is eliminated and one equation in x is left.

To solve $y = a_1x^2 + b_1x + c_1$, $y = a_2x^2 + b_2x + c_2$**:** (1) set the right sides equal (or subtract the equations); (2) collect all terms on one side and solve for x (factor, or use the quadratic formula); (3) substitute each x into the simpler original equation to find y; (4) verify in the **other** equation and write the ordered pairs.

Example 3

Solve each system algebraically and verify one solution.

(a) $y = x^2 - 2x - 4$
$y = -x^2 + 4x + 4$

(b) $y = 3x^2 + 2x - 1$
$y = x^2 - 3x + 2$

Example 4

Consider the system $y = x^2 - 3$ and $y = -2x^2 + 6x + 1$.

(a) Use a graphing calculator to solve the system, to the nearest tenth. State a window that shows both points.

(b) Solve the system algebraically to get the exact solutions.

3. When the x^2-Terms Cancel

If the two equations have the same x^2-coefficient, eliminating y also eliminates x^2, and what is left is **linear**:

- a linear equation $mx = n$ with $m \neq 0$: exactly **one** solution;
- a false statement such as $0 = 5$: **no** solution (one parabola is a vertical translation of the other);
- a true statement $0 = 0$: **infinitely many** solutions (the equations describe the same parabola).

Example 5

Solve each system, or explain why it has no solution or infinitely many solutions.

(a) $y = x^2 + 4x + 1$
$y = x^2 - 2x + 7$

(b) $y = 2x^2 - 3x + 1$
$y = 2x^2 - 3x - 4$

(c) $y = 2(x - 1)^2 + 3$
$y = 2x^2 - 4x + 5$

4. The Number of Solutions

When the x^2-terms do not cancel, the equation left after eliminating y is quadratic, and its discriminant counts the solutions.

After eliminating y	**The two parabolas**	**Solutions**
quadratic, $b^2 - 4ac > 0$	cross at two points	two
quadratic, $b^2 - 4ac = 0$	touch at one point	one
quadratic, $b^2 - 4ac < 0$	do not meet	none
linear, $mx = n$ $(m \neq 0)$	cross at one point	one
$0 = n$ $(n \neq 0)$	are vertical translations of each other	none
$0 = 0$	are the same parabola	infinitely many

Example 6

Without solving, determine the number of solutions of each system.

(a) $y = x^2 + 2x + 3,\ \ y = -x^2 - 2x + 1$

(b) $y = x^2 + 1,\ \ y = -x^2 + 2x - 2$

(c) $y = 2x^2 - 5,\ \ y = x^2 + x + 1$

(d) $y = -x^2 + 5x,\ \ y = -x^2 + 5x + 2$

Example 7

For which values of k does the system $y = x^2 + k$ and $y = -x^2 + 4x$ have exactly one solution? two solutions? no solution? Determine the solution in the one-solution case.

5. Fractional and Decimal Coefficients

Clear fractions or decimals **after** eliminating y: multiply the resulting equation by a number that makes every coefficient an integer. Then factor or use the formula as usual. In a context, check that each solution makes sense.

Example 8

(a) Solve $y = \frac{1}{2}x^2 - 2x + 1$ and $y = -\frac{1}{4}x^2 + x + 10$.

(b) Two water jets in a fountain follow the paths $h = -x^2 + 6x$ and $h = -0.5x^2 + 2x + 2$, where x is the horizontal distance and $h \geq 0$ the height, both in metres. Where do the jets cross? Answer to the nearest hundredth.

Lesson 3: Problems Involving Systems

Modelling situations with linear–quadratic and quadratic–quadratic systems, solving them, and interpreting the solutions in context.

1. Modelling with a System of Equations

Many problems have two unknown quantities linked by two conditions. Each condition gives one equation, and together the equations form a **system**. When one equation is linear and the other quadratic, it is a **linear–quadratic system**; when both are quadratic, it is a **quadratic–quadratic system**. Every solution (x, y) of the system must fit *both* conditions.

Solving a problem with a system: (1) define the variables, with units; (2) write one equation for each condition; (3) solve by substitution or elimination (or graph both equations and use *intersect*); (4) check each solution in the *original* conditions and reject any that make no sense (a negative length, a time before the start, a count that is not a whole number); (5) answer the question in a sentence, with units.

Example 1

Write a system of equations and solve it.

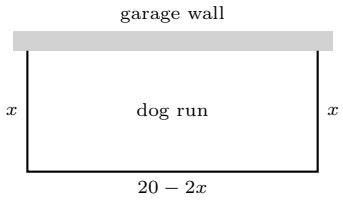
(a) The difference of two numbers is 4. The larger number plus the square of the smaller number is 46. Determine the numbers.

(b) Two positive integers have a sum of 15, and the sum of their squares is 117. Determine the integers.

Example 2

A rectangular dog run is built against the wall of a garage, so fencing is needed on only three sides. The width x is measured out from the wall, and 20 m of fencing is used.

(a) Show that the area is $A = 20x - 2x^2$.



(b) The run must have an area of 48 m^2. Solve the linear–quadratic system $y = 20x - 2x^2$, $y = 48$ to determine the possible dimensions.

(c) Could the run have an area of 60 m^2? Explain using the system $y = 20x - 2x^2$, $y = 60$.

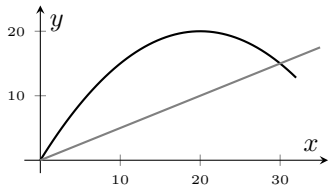
2. Linear–Quadratic Models

A straight path or edge (a hillside, a road, a ramp, a cost that grows at a constant rate) is modelled by a line; a curved path (a thrown object, an arch) by a parabola. Their points of intersection are where the two meet. Substitution usually leads to a quadratic equation; factor it or use the quadratic formula.

A **circle** centred at the origin with radius r has equation $x^2 + y^2 = r^2$ (every point is r units from $(0, 0)$, by the distance formula). It is also quadratic, and it is solved with a line the same way.

Example 3

A golf ball is hit from a tee at the bottom of a hill. Its path is $y = -0.05x^2 + 2x$ and the hillside is the line $y = 0.5x$, where x is the horizontal distance from the tee and y is the height above the tee, both in metres.



(a) Where does the ball land on the hill?

(b) How far up the slope from the tee does it land, to the nearest tenth of a metre?

(c) What is the greatest vertical height of the ball above the hillside?

Example 4

A radar station at the origin detects aircraft within 10 km, so its range is the circle $x^2 + y^2 = 100$ (units in kilometres). A plane flies along the line $y = x + 2$.

(a) Determine the points where the plane enters and leaves the radar range.

(b) How far does the plane fly inside the range, to the nearest tenth of a kilometre?

(c) Would a plane flying along $y = x + 15$ be detected? Explain.

3. Quadratic–Quadratic Models

Two objects moving under gravity, two curved paths, or a revenue and a cost that are both quadratic give a quadratic–quadratic system. Set the two expressions equal (or subtract the equations) and solve. If the squared terms are the same, they cancel and there is at most one solution (unless the two equations are the same).

Example 5

At the same moment, Ava throws a ball upward from a balcony and Ben throws one upward from the ground. After t seconds the heights, in metres, are

$$\text{Ava: } h = -5t^2 + 10t + 15 \qquad \text{Ben: } h = -5t^2 + 20t$$

(a) When are the two balls at the same height, and what is that height?

(b) Which ball lands first, and how much sooner?

Example 6

A company makes x hundred phone cases. Its revenue and its cost, both in hundreds of dollars, are

$$R = -2x^2 + 60x \qquad C = x^2 + 6x + 135$$

(a) The company **breaks even** when $R = C$. Determine the break-even quantities.

(b) For what quantities does the company make a profit?

(c) Profit is $P = R - C$. Determine the greatest profit and the quantity that gives it.

4. Using the Discriminant: Tangents and Unknown Constants

Substituting one equation into the other gives a quadratic equation $ax^2 + bx + c = 0$. Its discriminant $b^2 - 4ac$ counts the solutions of the system.

$b^2 - 4ac > 0$: two solutions (the graphs cross twice) $\quad b^2 - 4ac = 0$: one solution (the line is **tangent** to the parabola: it just touches it) $\quad b^2 - 4ac < 0$: no solution.

When one equation contains an unknown constant such as k, write the discriminant in terms of k and use the condition you need.

Example 7

(a) Determine the values of k for which the line $y = kx - 3$ is tangent to the parabola $y = x^2 + 1$, and the point of tangency for each.

(b) For what values of n does the system $y = x^2 - 4x + 7$ and $y = 2x + n$ have two solutions?

5. Finding a Model from Data

If a model has two unknown parameters, two known data points give two equations: a system. For $h = a(t-p)^2 + q$ with a known, subtracting the two equations eliminates q and leaves an equation in p only, because the p^2 terms cancel.

Example 8

A ball is thrown upward from a roof. Its height in metres after t seconds is $h = -5(t-p)^2 + q$. After 1 s the ball is 30 m above the ground, and after 4 s it is 45 m above the ground.

(a) Write a system of two equations in p and q, and solve it.

(b) How high is the roof, what is the maximum height, and when does the ball hit the ground (to the nearest hundredth of a second)?

Lesson 4: Linear Inequalities in Two Variables

Graphing the solution region of a linear inequality with a boundary line and a test point, writing an inequality from its graph, and using inequalities in context.

1. Review: Linear Inequalities in One Variable

An **inequality** uses one of the symbols $<$, $\leq$, $>$, $\geq$. Solve a linear inequality like an equation, with one extra rule.

Multiplying or dividing both sides by a **negative** number **reverses** the inequality sign: $-2x \geq 12 \Rightarrow x \leq -6$.

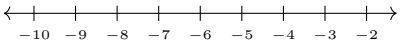
Check with a **test value**: a number in the solution should make the original inequality true, and one outside it false. On a number line, an **open circle** means the endpoint is not included ($<$, $>$) and a **closed dot** means it is ($\leq$, $\geq$). In set notation, $x \leq -6$ is written $\{x \mid x \leq -6, x \in \mathbb{R}\}$. Unless told otherwise, variables are real numbers.

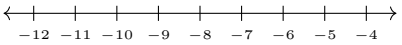
Example 1

Solve, check with a test value, graph the solution on the number line and write it in set notation.

(a) $3 - 2(x + 4) \geq 7$

(b) $\dfrac{x-1}{3} < \dfrac{2x+5}{4}$





2. Solutions of an Inequality in Two Variables

In a **linear inequality in two variables**, such as $2x - y < 4$, a solution is an ordered pair (x, y) that makes the inequality true. There are infinitely many solutions; together they fill a **solution region**. The line $2x - y = 4$, the **boundary line**, splits the plane into two **half-planes**, and the solution region is one of them.

Boundary: $\leq$ or $\geq$: **solid** line (points on the line are solutions) $<$ or $>$: **dashed** line (points on the line are not solutions)

Example 2

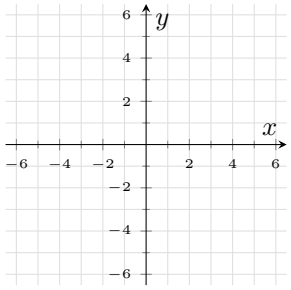
(a) Determine whether each point is a solution of $2x - y < 4$.

(i) $(0, 0)$ _____ (ii) $(3, 1)$ _____

(iii) $(1, -3)$ _____ (iv) $(2, 0)$ _____

(v) $(-2, 3)$ _____

(b) Graph the boundary line $2x - y = 4$ (should it be solid or dashed?), plot the five points, and shade the side of the line that holds the solutions.



3. Graphing a Linear Inequality

To graph a linear inequality: (1) graph the boundary line by replacing the sign with $=$ (use $y = mx + b$, or the two intercepts); (2) make it solid for $\leq$, $\geq$ or dashed for $<$, $>$; (3) choose a **test point** not on the line ($(0, 0)$ is easiest unless the line passes through it) and substitute it into the inequality; (4) if it is true, shade the side with the test point; if false, shade the other side.

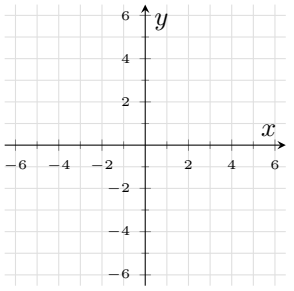
Shortcut: once the inequality is solved for y, $y > mx + b$ (or $\geq$) is the region **above** the line and $y < mx + b$ (or $\leq$) is **below**. Remember to reverse the sign if you divide by a negative while isolating y. Use a test point to confirm.

Graphing calculator (optional): solve for y, enter $mx + b$ in `Y=`, move the cursor to the left of `Y1` and press `ENTER` until the shade-above or shade-below symbol shows, then `GRAPH`. The calculator does not show solid or dashed lines, so you must decide that yourself.

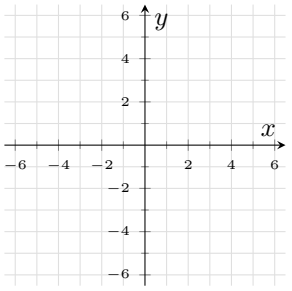
Example 3

Graph each inequality.

(a) $y < 2x - 1$



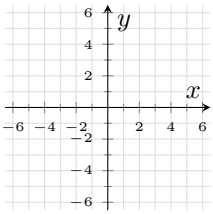
(b) $y \geq -\frac{2}{3}x + 3$



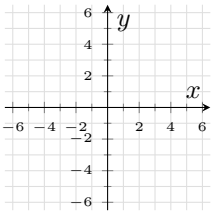
Example 4

Graph each inequality. Use intercepts, or solve for y first.

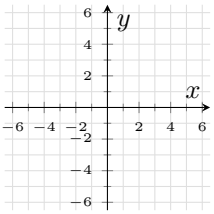
(a) $4x - 3y \geq 12$



(b) $2x + 5y < 0$



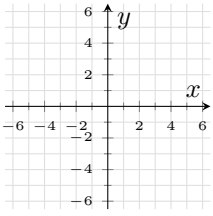
(c) $5x + 2y - 10 \leq 0$



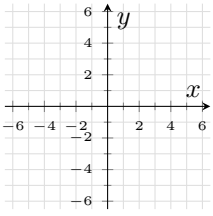
Example 5

Horizontal and vertical boundaries: $x > a$ is the half-plane to the right of the vertical line $x = a$, and $y \leq b$ is the half-plane below the horizontal line $y = b$. Graph each inequality.

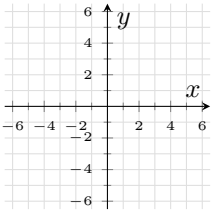
(a) $x > -2$



(b) $y + 1 \leq 0$



(c) $-3 \leq y < 2$



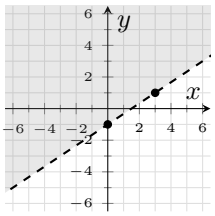
4. Writing an Inequality from Its Graph

(1) Write the equation of the boundary line (from its slope and y-intercept, two points, or the intercepts). (2) Dashed $\Rightarrow <$ or $>$; solid $\Rightarrow \leq$ or $\geq$. (3) Substitute a point from the shaded region to choose the direction of the sign (or use above/below when the line is written $y = mx + b$).

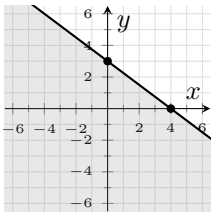
Example 6

Write the inequality whose solution region is shown.

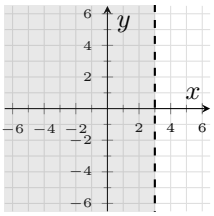
(a)



(b)



(c)



5. Inequalities in Context

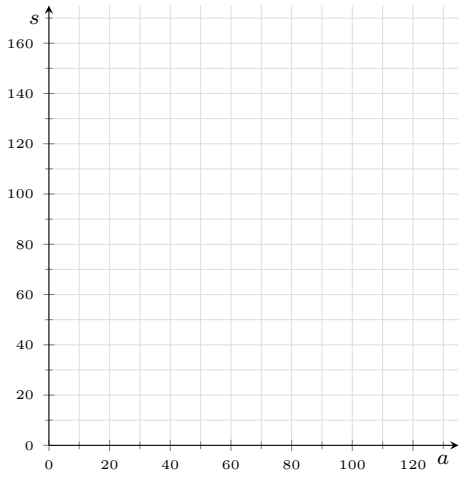
In a problem, define the variables, then write the inequality from the words: "at least" means $\geq$, "at most" or "no more than" means $\leq$, "more than" means $>$, "less than" means $<$. Add the **restrictions** the situation needs: quantities that cannot be negative give $x \geq 0$ and $y \geq 0$ (the graph stays in the first quadrant), and counts of people or objects must be whole numbers, so only the points with whole-number coordinates in the region are real answers. When two conditions must hold at once, graph both inequalities; the solution region is where the shadings overlap.

Example 7

A school play sells adult tickets for \$15 and student tickets for \$10. To cover its costs it must take in at least \$1500. Let a be the number of adult tickets and s the number of student tickets.

(a) Write an inequality, with restrictions, for the situation.

(b) Graph the solution region.



(c) Are 60 adult and 50 student tickets enough?

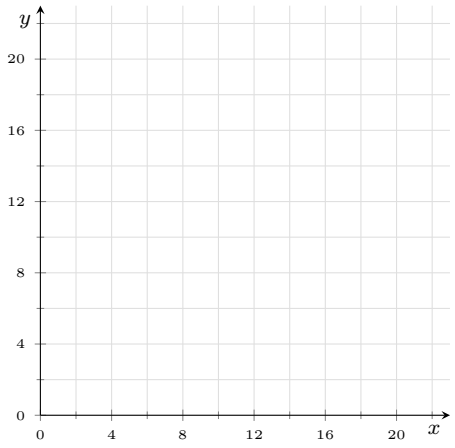
(d) If 40 adult tickets are sold, what is the least number of student tickets needed?

Example 8

Jordan earns \$20 per hour tutoring (x hours) and \$15 per hour lifeguarding (y hours). He can work at most 20 hours a week and wants to earn at least \$330.

(a) Write a system of inequalities, with restrictions.

(b) Graph the solution region.



(c) Can he work 10 hours tutoring and 8 hours lifeguarding?

(d) What is the least number of tutoring hours he can work and still meet both conditions?

Lesson 5: Quadratic Inequalities in Two Variables

Graphing the region above or below a parabola with a solid or dashed boundary and a test point, writing an inequality from its graph, and using the region in a context.

1. Solutions of a Quadratic Inequality in Two Variables

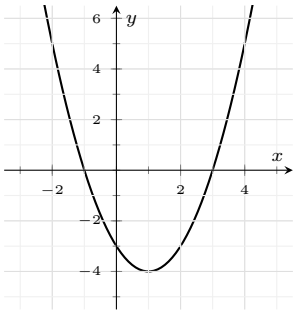
A **quadratic inequality in two variables** has one of the forms

$$y < ax^2 + bx + c, \qquad y \le ax^2 + bx + c, \qquad y > ax^2 + bx + c, \qquad y \ge ax^2 + bx + c, \qquad a \ne 0.$$

A **solution** is an ordered pair (x, y) that makes the inequality true. There are infinitely many, so we show them as a shaded **solution region**. As with lines in Lesson 4, the graph of the related equation $y = ax^2 + bx + c$ is the **boundary**. A parabola splits the plane into two regions: the region **inside** the parabola (the side it opens toward) and the region **outside** it. Every point of one region is a solution, and no point of the other is.

Example 1

Consider the inequality $y \geq x^2 - 2x - 3$. The boundary $y = x^2 - 2x - 3$ is graphed.



(a) For each point, find $x^2 - 2x - 3$ and decide whether the point is a solution. Plot the points, marking each solution.

Point	$x^2 - 2x - 3$	Is $y \geq x^2 - 2x - 3$?
$(0, 0)$		
$(4, 1)$		
$(1, -5)$		
$(-2, 5)$		
$(2, 1)$		
$(0, -4)$		
$(3, 0)$		

(b) Describe where the solutions lie.

(c) Which of the points are solutions of $y > x^2 - 2x - 3$? Explain the difference.

Boundary: solid parabola for $\leq$ or $\geq$ (points on it are solutions); dashed parabola for $<$ or $>$ (they are not).
Region (with y alone on the left): $y > \ldots$ or $y \geq \ldots$ is the region **above** the parabola; $y < \ldots$ or $y \leq \ldots$ is the region **below** it.
For a parabola that opens up, "above" is inside; for one that opens down, "above" is outside.

2. Graphing the Solution Region

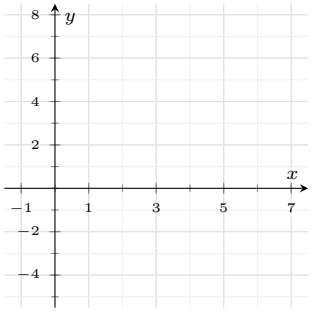
To graph a quadratic inequality in two variables:
(1) Graph the boundary $y = ax^2 + bx + c$: find the vertex, the x-intercepts (factor) and the y-intercept, and use symmetry. Draw it solid for $\leq, \geq$ and dashed for $<, >$.
(2) Choose a **test point** that is not on the parabola ($(0, 0)$ is easiest when it is not on it) and substitute it into the inequality.
(3) If the test point makes the inequality true, shade the region (inside or outside) that contains it; if false, shade the other region.

One test point is enough, because every point on the same side of the parabola gives the same answer. Use the test point to check the "above/below" rule, not instead of thinking about it.

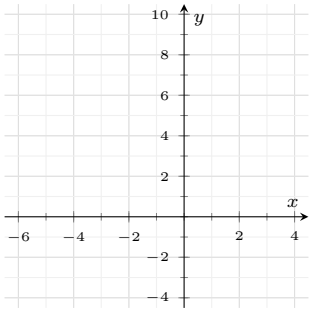
Example 2

Graph the solution region of each inequality.

(a) $y \leq x^2 - 6x + 5$



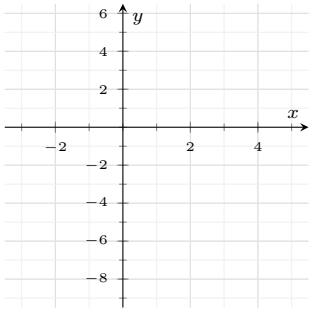
(b) $y > -x^2 - 2x + 8$



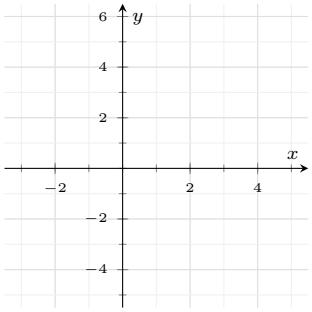
Example 3

Graph the solution region of each inequality. Use the form the equation is given in.

(a) $y \geq 2(x-1)^2 - 8$



(b) $y < -\frac{1}{2}(x+2)(x-4)$



3. Rearranging First, and Choosing a Test Point

If y is not alone on the left, solve for y first, as you would solve an equation, with one change: **multiplying or dividing both sides by a negative number reverses the inequality sign**. For example, $-y > 3$ becomes $y < -3$. (Dividing by a positive number keeps the sign.)

When $c = 0$ the parabola passes through $(0, 0)$, so $(0, 0)$ cannot be the test point: it gives an equality, which tells you nothing about the regions. Use a point such as $(1, 0)$, $(0, 1)$ or a point straight above the vertex.

Example 4

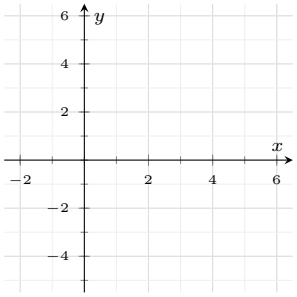
Write each inequality with y alone on the left. State whether the boundary is solid or dashed and whether the region is above or below the parabola.

(a) $y - 3 > x^2 - 4x$ (b) $x^2 - y \geq 4x$ (c) $4 - 2y < 2x^2 + 6x$

Example 5

(a) Explain why $(0, 0)$ cannot be used as the test point for $y \leq x^2 - 4x$, the inequality from Example 4(b).

(b) Graph the solution region of $y \leq x^2 - 4x$.



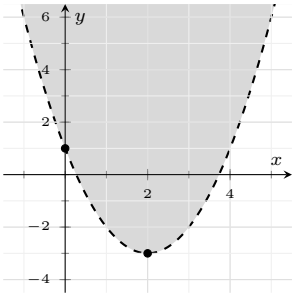
4. Writing an Inequality from a Graph

To write the inequality for a shaded region:
(1) Write the equation of the boundary parabola: from the vertex (p, q) and a point, use $y = a(x - p)^2 + q$; from the x-intercepts r, s and a point, use $y = a(x - r)(x - s)$. Substitute the point to find a.
(2) Choose the sign: shaded above $\Rightarrow >$ or $\geq$; shaded below $\Rightarrow <$ or $\leq$; solid $\Rightarrow$ include "="; dashed $\Rightarrow$ do not.
(3) Check with a point in the shaded region.

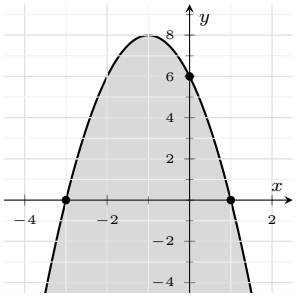
Example 6

Write the inequality whose solution region is shown, in general form.

(a) Vertex $(2, -3)$; the point $(0, 1)$ is on the boundary.



(b) x-intercepts -3 and 1; y-intercept 6.



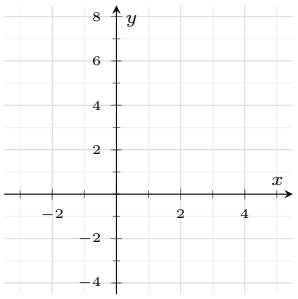
5. Using a Graphing Calculator, and Regions in Context

A graphing calculator shades a region quickly, which helps when the intercepts are not whole numbers.

TI-84 Plus CE: put y alone on the left; in [Y=] enter the right side as Y_1; move the cursor to the far left of Y_1 and press [ENTER] until the shade-above icon (for $>$, $\geq$) or the shade-below icon (for $<$, $\leq$) shows; press [GRAPH]. Use 2nd [CALC] zero for the x-intercepts.
(On the Casio, choose the graph type $Y>$, $Y<$, $Y\geq$ or $Y\leq$ for the function.)
The calculator does not show solid or dashed boundaries: decide that yourself when you sketch.

Example 7

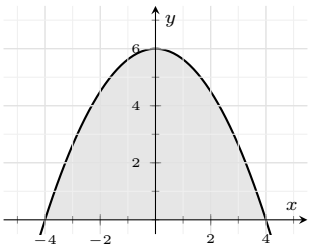
Use a graphing calculator to graph $y < -x^2 + 2x + 6$. Sketch the region on the grid, and label the intercepts and the vertex (to the nearest tenth).



Regions bounded by parabolas describe real situations: the space under an arch, under the path of a ball or a jet of water, or inside a reflector. A point (x, y) is inside the region when it satisfies the inequality, and an object fits when all its corner points do.

Example 8

The opening of a tunnel is a parabolic arch. With the origin at the centre of the road, the arch is $y = -0.375x^2 + 6$, where x and y are in metres.
(a) Write the inequalities for the points on or under the arch and on or above the road. How wide is the road under the arch?



(b) A truck 3.2 m wide and 4.5 m tall drives along the centre of the road. Will it fit? Show the test.

(c) How tall can a 4 m wide truck in the centre be?

Lesson 6: Quadratic Inequalities in One Variable: Roots and Test Points

Solving a quadratic inequality in one variable from a graph and from its roots with a test point in each interval, and writing the solution in set notation and on a number line.

1. Quadratic Inequalities and Their Graphs

A **quadratic inequality in one variable** has one of the forms

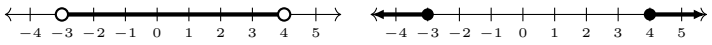
$$ax^2 + bx + c < 0, \qquad ax^2 + bx + c \le 0, \qquad ax^2 + bx + c > 0, \qquad ax^2 + bx + c \ge 0, \qquad a \ne 0.$$

Its **solution** is the set of all x-values that make it true, usually an interval or two. The graph of the related function $y = ax^2 + bx + c$ shows them: $ax^2 + bx + c > 0$ exactly where the graph is **above** the x-axis, and $ax^2 + bx + c < 0$ where it is **below**. The x-intercepts (the roots of $ax^2 + bx + c = 0$) are where it changes from one to the other.

From the graph of $y = ax^2 + bx + c$: > 0: the x-values where the graph is above the x-axis; < 0: where it is below; $\ge$ or $\le$: include the x-intercepts.

Writing the solution: between the roots: $\{x \mid -3 < x < 4, x \in \mathbb{R}\}$; outside the roots: $\{x \mid x \le -3 \text{ or } x \ge 4, x \in \mathbb{R}\}$.

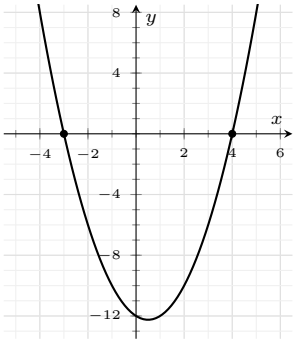
On a number line, an **open dot** means the endpoint is not included ($<$, $>$) and a **closed dot** means it is ($\le$, $\ge$).



Two separate pieces are joined by "or" (an x cannot be in both). Never write $4 < x < -3$: it says x is bigger than 4 and less than -3 at once.

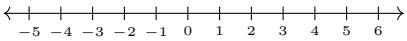
Example 1

The graph of $y = x^2 - x - 12$ is shown. Use it to solve each equation or inequality. Show (b) and (c) on a number line.

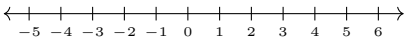


(a) $x^2 - x - 12 = 0$

(b) $x^2 - x - 12 < 0$

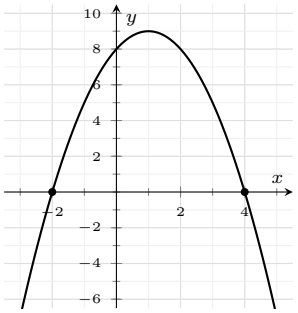


(c) $x^2 - x - 12 \ge 0$



Example 2

The graph of $y = -x^2 + 2x + 8$ is shown. Use it to solve each inequality.



(a) $-x^2 + 2x + 8 > 0$

(b) $-x^2 + 2x + 8 \le 0$

(c) $-x^2 + 2x + 8 > 5$ (Draw the line $y = 5$.)

2. Solving with a Graphing Calculator

When the roots are not whole numbers, a graphing calculator reads them quickly.

(1) Move every term to one side: $ax^2 + bx + c \square 0$. (2) Graph $Y_1 = ax^2 + bx + c$. (3) Use 2nd [CALC] zero to find each x-intercept. (4) Read where the graph is above (> 0) or below (< 0) the x-axis.

Or, without rearranging: graph Y_1 = left side and Y_2 = right side, use intersect, and read where Y_1 is above or below Y_2.

The quadratic formula gives the same roots exactly: use it when the question asks for exact values.

Example 3

Solve each inequality with a graphing calculator, to the nearest tenth. Then give the exact solution.

(a) $x^2 - 4x - 3 > 0$

(b) $2x^2 \leq 5x + 4$

3. Roots and Test Points

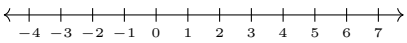
A graph is not needed. The graph of $y = ax^2 + bx + c$ can only cross the x-axis at a root, so the expression $ax^2 + bx + c$ can only change sign at a root. The roots split the number line into **intervals**, and on each interval the expression keeps one sign. One **test point** in each interval tells you that sign.

Roots and test points:
(1) Write the inequality with 0 on one side. (2) Solve the related equation $ax^2 + bx + c = 0$ (factor, or use the quadratic formula). These roots are the boundary points. (3) Mark the roots on a number line; they make two or three intervals. (4) Substitute one test point from each interval into the inequality. (5) The solution is every interval whose test point makes it true; include the roots for $\leq$ or $\geq$, exclude them for $<$ or $>$.

Example 4

Solve $x^2 - 3x - 10 > 0$ using roots and test points.

Interval			
Test point			
Value of $x^2 - 3x - 10$			
Is it > 0?			

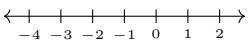


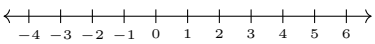
Example 5

Solve using roots and test points. Show each solution on a number line.

(a) $2x^2 + 5x \leq 3$

(b) $15 + 2x - x^2 < 0$





In (b) you could first multiply by -1 to get $x^2 - 2x - 15 > 0$ (the sign reverses); the test points still check the original inequality, which is the safest habit.

Example 6

Expand or rearrange first, then solve.

(a) $x(x+4) \geq 5$

(b) $(x-2)^2 < 3x - 2$

Another algebraic method looks at the signs of the factors (case analysis); Lesson 7 develops it into sign charts.

4. When There Are One or No Roots

If the related equation has a double root, or no real roots (discriminant $b^2 - 4ac < 0$), the graph touches the x-axis once or never. Then the expression has the same sign everywhere (except 0 at a double root), and one test point decides everything. The solution can be all real numbers, one number, all real numbers but one, or no real numbers at all.

Example 7

Solve each inequality.

(a) $x^2 - 8x + 16 > 0$

(b) $x^2 - 8x + 16 \leq 0$

(c) $x^2 + 2x + 5 < 0$

(d) $x^2 + 2x + 5 > 0$

5. Applications

Quadratic inequalities answer "for which values" questions: which values of a parameter give two, one or no roots (the discriminant from Unit 5), and which measurements keep a quantity in a range. In a context, also apply the restrictions of the situation (a length is positive, a count is a whole number).

Example 8

(a) For what values of k does $x^2 + kx + 9 = 0$ have two distinct real roots? For what values does it have no real roots?

(b) The length of a rectangle is 4 cm more than its width. For what widths is the area at most 21 cm^2?

Lesson 7: Quadratic Inequalities in One Variable: Sign Analysis

Solving a quadratic inequality from the signs of its factors, by cases and with a sign chart, and using it to answer questions about heights, profits and coefficients.

1. The Sign of a Product

Lesson 6 solved quadratic inequalities with roots and test points. Here the inequality is solved from its **factors**. Once one side is 0 and the other side is factored, only the **signs** of the factors matter.

Signs of a product. $AB > 0$ when A and B have the **same** sign (both positive or both negative). $AB < 0$ when A and B have **opposite** signs. $AB = 0$ when $A = 0$ or $B = 0$.

Case analysis turns this into pairs of linear inequalities. Within a case both inequalities must hold ("and": keep the overlap); the solution is the result of Case 1 **or** Case 2. A case can have no solution. Solutions are written in set notation, for example $\{x \mid x < -2 \text{ or } x > 3, x \in \mathbb{R}\}$, and can be shown on a number line (closed dot $\bullet$ for $\leq$ or $\geq$, open dot $\circ$ for $<$ or $>$).

Example 1

Solve by case analysis. Show each solution on a number line.

(a) $(x - 3)(x + 2) > 0$

(b) $x^2 + 4x - 21 \leq 0$

2. Sign Charts

Case analysis gets long. A **sign chart** does the same reasoning in a table. A linear factor such as $x - 5$ is 0 at its zero ($x = 5$), negative to the left of it and positive to the right (when the coefficient of x is positive). The zeros of the factors split the number line into intervals, and in each interval every factor keeps one sign.

Solving with a sign chart. (1) Move every term to one side so the other side is 0. (2) Factor fully. (3) Find the zero of each factor and write the zeros in order across the top of the chart, with an interval between and around them. (4) Fill in a row for each factor: $-$, 0 or $+$. (5) Product row: an even number of negative factors gives $+$, an odd number gives $-$. (6) Read the intervals whose sign matches the inequality (> 0: $+$; < 0: $-$). Include the zeros for $\leq$ or $\geq$; leave them out for $<$ or $>$.

Example 2

(a) Complete the sign chart and solve $x^2 - 3x - 10 > 0$.

x	$x < -2$	-2	$-2 < x < 5$	5	$x > 5$
$x + 2$					
$x - 5$					
Product					

(b) Use the same chart to solve $x^2 - 3x - 10 \leq 0$.

Example 3

Rearrange so one side is 0, then solve with a sign chart. Never divide both sides by an expression containing x, and do not set each factor equal to the number on the right side.

(a) $6x^2 + x \leq 2$

(b) $x(x + 6) < 16$

Example 4

When the leading coefficient is negative, either keep the negative constant as a row of the chart (it is negative everywhere), or multiply both sides by -1 and **reverse the inequality sign**. Solve $-x^2 + 2x + 15 \geq 0$ both ways.

(a) Keep -1 as a factor.

(b) Multiply by -1 first.

3. Squared Factors and Quadratics That Do Not Factor

A squared factor such as $(x-5)^2$ is never negative: it is 0 at $x = 5$ and positive everywhere else, so it **does not change sign**. If $ax^2 + bx + c$ has **no real zeros** (discriminant $b^2 - 4ac < 0$), its graph never crosses the x-axis, so it has one sign for every x: test any one value. If it has real zeros but does not factor, find the zeros with the quadratic formula and use them as the boundaries of the intervals.

Example 5

Solve.

(a) $x^2 - 10x + 25 > 0$

(b) $4x^2 + 12x + 9 \leq 0$

(c) $x^2 - 4x + 7 < 0$

(d) $x^2 - 2x - 4 \geq 0$ (exact, then to the nearest hundredth)

Example 6

(Extension.) A sign chart works for a product of any number of factors. Solve.

(a) $(x+2)(x-1)(x-4) \geq 0$

(b) $(x-2)^2(x+1) > 0$

4. Writing an Inequality from Its Solution

Work backwards. Each endpoint r gives a factor $(x-r)$; an endpoint $\frac{p}{q}$ gives $(qx-p)$. With a positive leading coefficient, the quadratic is **negative between** its zeros and **positive outside** them.

Solution **between** r_1 and r_2: $(x-r_1)(x-r_2) < 0$ (or ≤ 0). Solution **outside** ($x < r_1$ or $x > r_2$): $(x-r_1)(x-r_2) > 0$ (or ≥ 0). Closed dots (endpoints included) $\Rightarrow \leq$ or $\geq$; open dots $\Rightarrow <$ or $>$.

Example 7

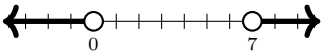
Write a quadratic inequality, in the form $ax^2+bx+c < 0$ (or $>, \leq, \geq$) with integer coefficients and $a > 0$, that has the given solution.

(a) $\{x \mid -4 < x < 3, x \in \mathbb{R}\}$

(b) $\{x \mid x \leq -6 \text{ or } x \geq -1, x \in \mathbb{R}\}$

(c) $\{x \mid -\frac{1}{2} \leq x \leq 3, x \in \mathbb{R}\}$

(d)



Multiplying by -1 and reversing the sign gives another correct inequality: (a) is also $-x^2-x+12 > 0$.

5. Problems

Write the inequality from the words ("at least" $\rightarrow \geq$, "more than" $\rightarrow >$, "at most" $\rightarrow \leq$), move every term to one side, solve, and answer in context. A time or a price cannot be negative.

Example 8

A model rocket is launched from the ground. Its height after t seconds is $h = 40t - 5t^2$ metres.

(a) When is the rocket at least 60 m high?

(b) For how many seconds is it at least 60 m high?

(c) For what times is the rocket in the air?

(d) Show that the rocket never reaches 90 m.

Example 9

A theatre group estimates that if a ticket costs x dollars, its profit for one show is $P = -20x^2 + 1000x - 8000$ dollars.

(a) For what ticket prices does the group make a profit?

(b) For what prices is the profit at least \$3000? Give the prices to the nearest cent.

Example 10

For what values of k does $x^2 + (k-2)x + 4 = 0$ have no real roots?

Lesson 8: Unit 8 Review

Systems of equations, inequalities in two variables and quadratic inequalities in one variable on one page, mixed examples that combine them, and a review assignment written like the unit test.

---

1. Unit Summary

Topic	Method	Example
Linear–quadratic system	Write the line as $y = \dots$ and substitute into the quadratic (or eliminate y). Solve the quadratic, find each y from the **linear** equation, verify in both. Graphically: graph both and use *intersect*.	$y = x^2 - 1$, $y = x + 1$ $x^2 - x - 2 = 0$, $x = 2$ or -1 Solutions $(2, 3)$ and $(-1, 0)$
Quadratic–quadratic system	Set the right sides equal, collect terms on one side and solve. If the x^2-terms cancel, the equation is linear.	$y = x^2 + x$, $y = 2x^2 - x - 3$ $x^2 - 2x - 3 = 0$, $x = 3$ or -1 Solutions $(3, 12)$ and $(-1, 0)$
Linear inequality in two variables	Graph the boundary line: **solid** for $\leq$, $\geq$; **dashed** for $<$, $>$. Substitute a **test point** not on the line ($(0, 0)$ if possible); shade its side if true, the other side if false.	$y < 2x + 1$: dashed line; $(0, 0)$: $0 < 1$ true, so shade the side with the origin (below).
Quadratic inequality in two variables	The same steps with a boundary parabola (vertex, intercepts). $y > \dots$ is above the parabola, $y < \dots$ below.	$y \geq x^2 - 4$: solid; $(0, 0)$: $0 \geq -4$ true, so shade above (inside).
Quadratic inequality in one variable	One side 0, factor, find the roots. Use test points, a sign chart, or the graph (where it is above/below the x-axis). Answer in set notation.	$x^2 - x - 6 \leq 0$: $(x - 3)(x + 2) \leq 0$ $\{x \mid -2 \leq x \leq 3, x \in \mathbb{R}\}$

Number of solutions of a system. After eliminating y, the quadratic equation's discriminant tells how many times the graphs meet: $b^2 - 4ac > 0$: two solutions; $= 0$: one (a line meeting a parabola this way is **tangent** to it); < 0: none. Two parabolas can also be the same parabola: infinitely many solutions.

Solving and checking. A solution of a system is a point (x, y) on both graphs: check it in **both** equations. A solution of an inequality in two variables is a **region** of points; a solution of an inequality in one variable is a set of x-values on a number line. In a word problem, define the variables, reject values that make no sense (negative times or lengths, fractions of items) and answer in context.

Quadratic inequalities in one variable. $a > 0$: $ax^2 + bx + c < 0$ between the roots, > 0 outside them. Include the roots for $\leq$, $\geq$. Multiply by a negative number $\Rightarrow$ reverse the sign. No real roots ($b^2 - 4ac < 0$): the expression has one sign everywhere (test one value): all real numbers or no solution.

Common errors. Finding x and forgetting y. Substituting into the wrong equation without checking in the other. A dashed line for $\geq$. Using $(0, 0)$ as a test point when it is on the boundary. Setting each factor equal to the number on the right ($x(x + 6) < 16$ does not mean $x < 16$). Forgetting to reverse the sign when dividing by a negative. Writing "$x < \pm 3$" for $x^2 < 9$.

2. Systems of Equations

Example 1

(a) Solve the system $y = x^2 - 4x + 1$ and $y = 2x - 4$ algebraically, and verify.

(b) Determine k so that the line $y = 2x + k$ is tangent to $y = x^2 - 4x + 1$, and find the point of tangency.

Example 2

(a) Solve $y = 2x^2 + x - 3$ and $y = x^2 + 3x + 5$.

(b) How many solutions does the system $y = x^2 + 2$, $y = -x^2 + 2x$ have? Explain.

Example 3

A company's revenue and cost for making and selling x hundred items are $R = -2x^2 + 60x$ and $C = 10x + 200$, both in hundreds of dollars.

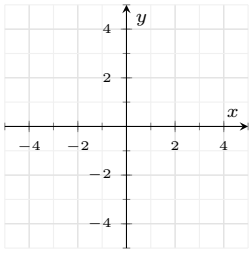
(a) Determine the break-even points (where $R = C$).

(b) For what production levels does the company make a profit?

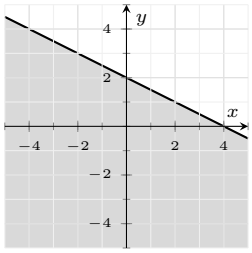
3. Inequalities in Two Variables

Example 4

(a) Graph $3x - 2y < 6$.

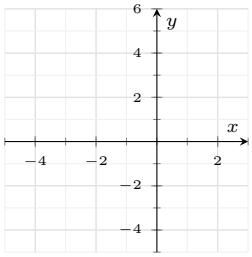


(b) The graph shows the solution region of a linear inequality. The boundary passes through $(0, 2)$ and $(4, 0)$. Write the inequality.



Example 5

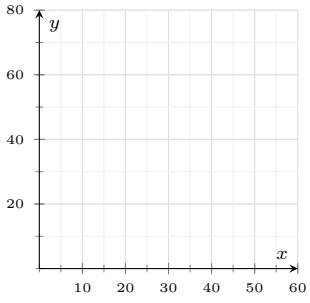
(a) Graph $y \geq x^2 + 2x - 3$.



(b) Is $(-2, 1)$ a solution? Is $(1, -1)$?

Example 6

A school play needs ticket sales of at least \$480. Adult tickets cost \$12 and student tickets cost \$8. Let x be the number of adult tickets and y the number of student tickets.
(a) Write an inequality, with any restrictions, and graph it.



(b) Do 20 adult and 25 student tickets reach the goal? Do 24 and 24?

4. Quadratic Inequalities in One Variable

Example 7

(a) Solve $2x^2 - x - 15 \geq 0$ with a sign chart.

(b) Use the graph in Example 5 to solve $x^2 + 2x - 3 < 0$. Explain.